

Chapter 01: Agents + Fundamentals

What Is an Agent — and Why Does It Matter Now?

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Review + Questions

Semester Plan

1. **Agents and their Fundamentals**
2. Perception, Memory, Planning & Learning
3. LLMs, RAG & Tool Use
4. Multi-Agent Systems
5. Agentic AI Engineering
6. Safety & Guardrails

The Three-Window Demo

Same task, three very different systems. What separates them?

1 *LLM Chatbot*

Answers politely,
explains options —
but does nothing.

2 *Travel Agent*

Searches, compares,
delivers a concrete
booking result.

3 *MARS Simulation*

Many simple agents,
emergent group
behavior visible.

Prompt: *"Book me a flight Hamburg → Berlin next Monday."*



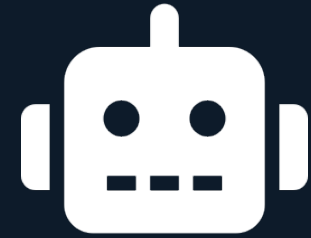
*What do these three systems
have in common — and what
separates them fundamentally?*

We will not resolve this now. The answer will emerge by the end of this session.

PART 1

What Is an Agent?

From everyday intuition to the formal definition by Russell & Norvig



From Intuition to Formal Definition

Everyday understanding

An agent is someone who acts on behalf of someone else
— a broker, a secret agent, a software bot.

Intelligence is always situated — it does not exist in a vacuum, but in relation to an environment.

Formal definition (Russell & Norvig)

"An agent is anything that can perceive its environment through sensors and acts upon that environment through actuators."

Key addition

An agent maximizes a performance measure — it acts purposefully, not randomly.

The PEAS Framework

The first tool for specifying what an agent must do — systematically.

Letter	Component	Guiding Question
P	Performance Measure	How does the agent know it succeeded?
E	Environment	In which world does it operate?
A	Actuators	How does it affect the world?
S	Sensors	How does it perceive the world?

Source: Russell & Norvig, Artificial Intelligence: A Modern Approach (1995)

PEAS in Action: Four Examples

	Spam Filter	Tesla Autopilot	ChatGPT	Travel Agent
P	Accuracy, low false positives	Safety, comfort, efficiency	User satisfaction (?)	Cost, time, user preference
E	Email inbox	Roads, traffic, weather	Conversation (?)	Web, APIs, calendars
A	Move, flag, delete	Steering, braking, signals	Text output only (?)	Search, book, notify
S	Email headers, content	Cameras, LIDAR, GPS	Text input only (?)	Web data, user input

(?) marks where the classification becomes debatable. Is a pure LLM really an agent?

Is ChatGPT an Agent?

The PEAS analysis reveals a structural gap — not just a limitation.

What a pure LLM lacks

- No real actuators — text output only
- No persistent environment
- No memory beyond the context window
- No clear performance measure
- No ability to act on the world

The key insight

An LLM is a powerful reasoning engine — but it is not situated in an environment.

To become an agent, it needs:

- Tools (actuators)
- Memory (persistence)
- Grounding (sensors)

This is exactly what Chapters 02 + 03 will address.

PART 2

Agent Taxonomy

Five types as an evolutionary story — each generation fixes the previous one's weakness.



Taxonomy: Reflex Agents

① Simple Reflex Agent

"If dirty → suck. If clean → move."

Reacts directly to current perception.
No memory, no goals, no model of the world.

Limitation:

Only works in fully observable, predictable environments.

→ fixes

② Model-Based Reflex Agent

"Keeps an internal world model."

Maintains state about what it cannot currently see. Tracks how the world changes.

Limitation:

Still only reacts — it does not plan ahead.

Taxonomy: Deliberative Agents

③ Goal-Based Agent

"Knows where it wants to go."

Searches for action sequences that lead to a goal state. Can plan ahead.

Limitation:

Goals can be equally valid or even conflicting — how to choose?

→ fixes

④ Utility-Based Agent

"Evaluates and picks the best option."

Rates states with a utility function. Selects the action that maximizes expected utility.

Limitation:

Can optimize — but can it learn from experience?

Taxonomy: The Learning Agent

⑤ Learning Agent

Improves through experience. Has a learning component, a critic, and a performance element.
This is the most complete agent architecture in classical AI.

Learning Element

Makes improvements
based on feedback

Performance Element

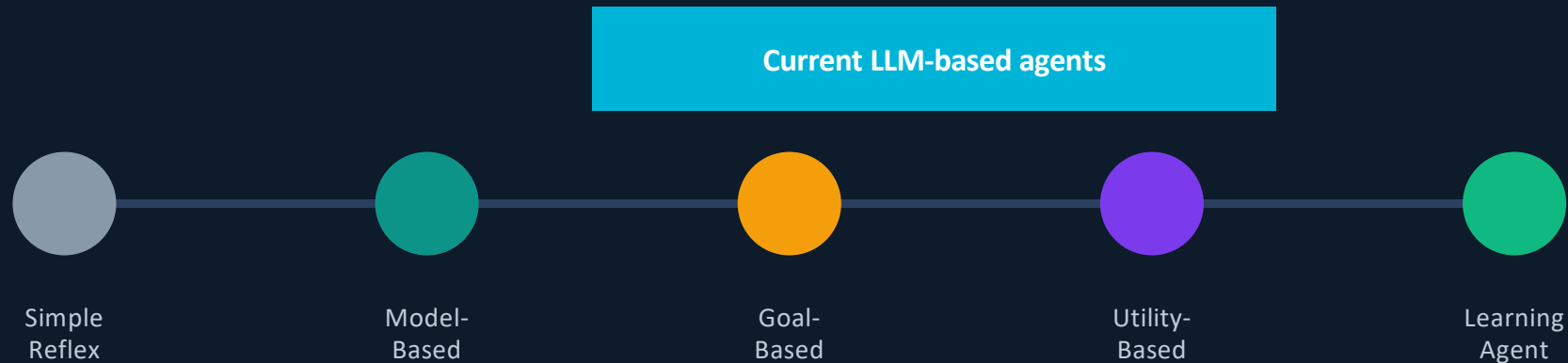
Selects actions
based on current knowledge

Critic

Evaluates outcomes
against a standard

Where Do LLM-Based Agents Fit?

A classification that reveals what is still missing.



What is still missing for true agency?

No persistent memory across sessions · No real-world perception · No continuous learning loop · No robust goal verification

Interlude: ABM vs. Agentic AI

The MARS simulation from the demo raises a question: both use "agents" — but they are fundamentally different.

Dimension	ABM (e.g. MARS)	Agentic AI
Purpose	Understand + explain	Complete tasks
Emergence	Desired — it is the insight	Undesired — it is a risk
Behavior	Deliberately simple	Deliberately complex

ABM asks: *"What emerges?"*

Agentic AI asks: *"What should emerge — and how do we prevent everything else?"*

PART 3

Environment Properties

Seven dimensions to classify any environment — and understand what the agent must handle.



Seven Environment Dimensions

Each dimension is a spectrum — not a binary category.

Dimension	Simpler ←		→ More Complex
Observability	Fully Observable	↔	Partially Observable
Outcome	Deterministic	↔	Stochastic
Episodes	Episodic	↔	Sequential
Dynamics	Static	↔	Dynamic
State Space	Discrete	↔	Continuous
Agents	Single-Agent	↔	Multi-Agent
Knowledge	Known	↔	Unknown

The more complex the environment dimensions, the higher the demands on the agent — and the harder the system is to build.

Classify These Systems

Where do these systems fall on the seven dimensions? Discuss edge cases.

Chess

Deterministic? Yes. Fully observable? Only if you ignore the opponent's thinking.

Uber Driver App

Dynamic, stochastic, partially observable. Multi-agent by nature.

Spam Filter

Mostly episodic — but is it really single-agent if spammers adapt?

MARS LaserTag

Simulated but complex: multi-agent, sequential, partially observable.

Travel Agent

Real-world APIs: dynamic, stochastic, partially observable, multi-agent.

PART 4

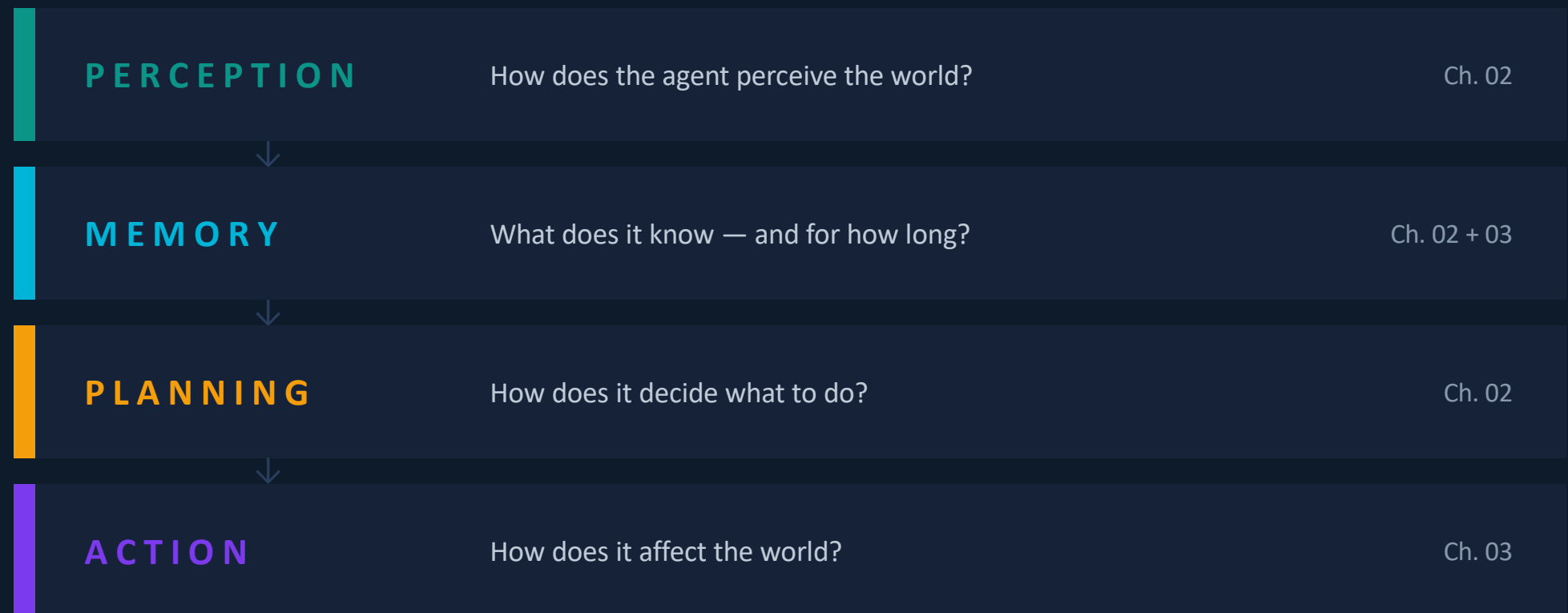
Cognitive Architecture



The four-layer model that serves as the analytical framework for the entire module.

The Four-Layer Model

Not an implementation detail — this is the analytical framework for the entire module.



Mapping the Three-Window Demo

The opening question now answers itself.

System	Perception	Memory	Planning	Action
ChatGPT (pure LLM)	Text input only	Context window only	No	Text output only
Travel Agent	Web, APIs, user input	Session + external tools	Yes (goal planning)	Search, book, notify
MARS LaserTag	Simulated sensors	None / rule-based	Reactive only	Simulated movement

The difference is not capability — it is architecture. Each system lacks different layers.

What Comes Next

Each following chapter fills one or more layers of the cognitive architecture.

Ch. 02 *Perception, Memory, Planning*

How agents perceive, remember, and decide — the cognitive core.

Ch. 03 *LLMs, RAG & Tool Use*

How language models close the gaps: RAG = Memory, Tools = Action.

Ch. 04 *Multi-Agent Systems*

How multiple agents organize, communicate, and coordinate.

Ch. 05 *Agentic AI Engineering*

How to build this in practice — frameworks, state, testing.

Ch. 06 *Safety & Guardrails*

When an agent must NOT act — and how to enforce it.

Key Takeaways

1

An agent is not just a program — it is a program in relation to its environment.

2

PEAS forces you to think about the goal before you think about the code.

3

The five agent types form an evolutionary chain — each fixing the previous one's weakness.

4

LLMs are powerful reasoning engines — but they are not agents by themselves.

5

The four-layer model (Perception → Memory → Planning → Action) is the analytical framework for this entire module.

Module Reflection

The Red Thread

- | | | |
|----|------------------------------|---|
| 01 | What is an agent? | → PEAS, Taxonomy, Cognitive Architecture |
| 02 | How does it perceive & plan? | → Perception, Memory, Planning |
| 03 | How does it speak? | → LLMs, RAG, Tool Use, React |
| 04 | How does it cooperate? | → Multi-Agent, Orchestrator/Worker |
| 05 | How is it built? | → Frameworks, State, Testing, Observability |
| 06 | How does it stay safe? | → Guardrails, HITL, Risk Analysis |

Next Session: Chapter 02

We now know what an agent is.

Next week we ask:

*How does an agent perceive the world,
what does it remember,
and how does it decide?*

Perception

Sensor types, state
representation, grounding

Memory

Short-term, long-term,
episodic memory types

Planning

Reactive vs. deliberative,
planning algorithms

References & Further Reading

Russell, S. & Norvig, P., "Artificial Intelligence: A Modern Approach" — Chapters 2.1–2.4 (HIBS)

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